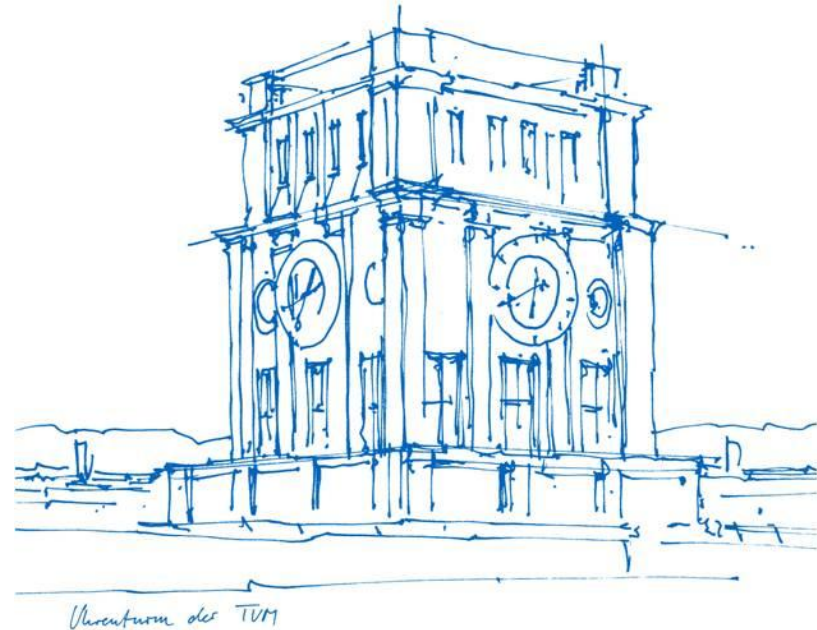


Group 2D - Dynamic Pricing

Team Members:

- Richard Horvath
- Adam Pasálek
- Lucas Krämer Sepúlveda
- Aijia Liu
- Zhixian Huang

Heilbronn, 16th July 2026



Intro

Business Problem: Fixed fares cannot respond to sudden changes in demand and driver availability.

Result: long waiting times, driver shortages, and lost revenue.

Dynamic pricing adjusts fares according to real-time market conditions:

- Higher demand → higher prices
- Lower driver supply → higher prices
- Balanced demand and supply → stable prices

Goal: Analyze demand, supply, booking time, location, and ride characteristics to identify surge-pricing conditions and build a predictive pricing model.

Context

The Dataset: Dynamic Pricing (1,000 records including driver/rider counts, location, and historical cost).

Core Challenge: Balancing rider demand with driver availability.

Business Risks: Failing to balance supply and demand leads to lost revenue, fixed price models fail to account for demand imbalances.

Objective: Address these critical inefficiencies to retain users and maximize profitability.

The Uber logo, consisting of the word "Uber" in a bold, black, sans-serif font.

Research Questions

The Operational View: Is the Pricing system efficient depending on the duration of the ride?

How does the demand for specific vehicle classes diverge from fleet availability across different times of the day?

The Algorithmic View: How can a predictive cost algorithm be leveraged to implement dynamic pricing, and what features have the most impact?

Understanding Data & Quality Inspection

Dataset Overview:

- 1000 Ride Records
- 10 Original Variables
 - 6 Numerical
 - 4 Categorical
- Target:
 - [Historical_Cost_of_Ride](#)

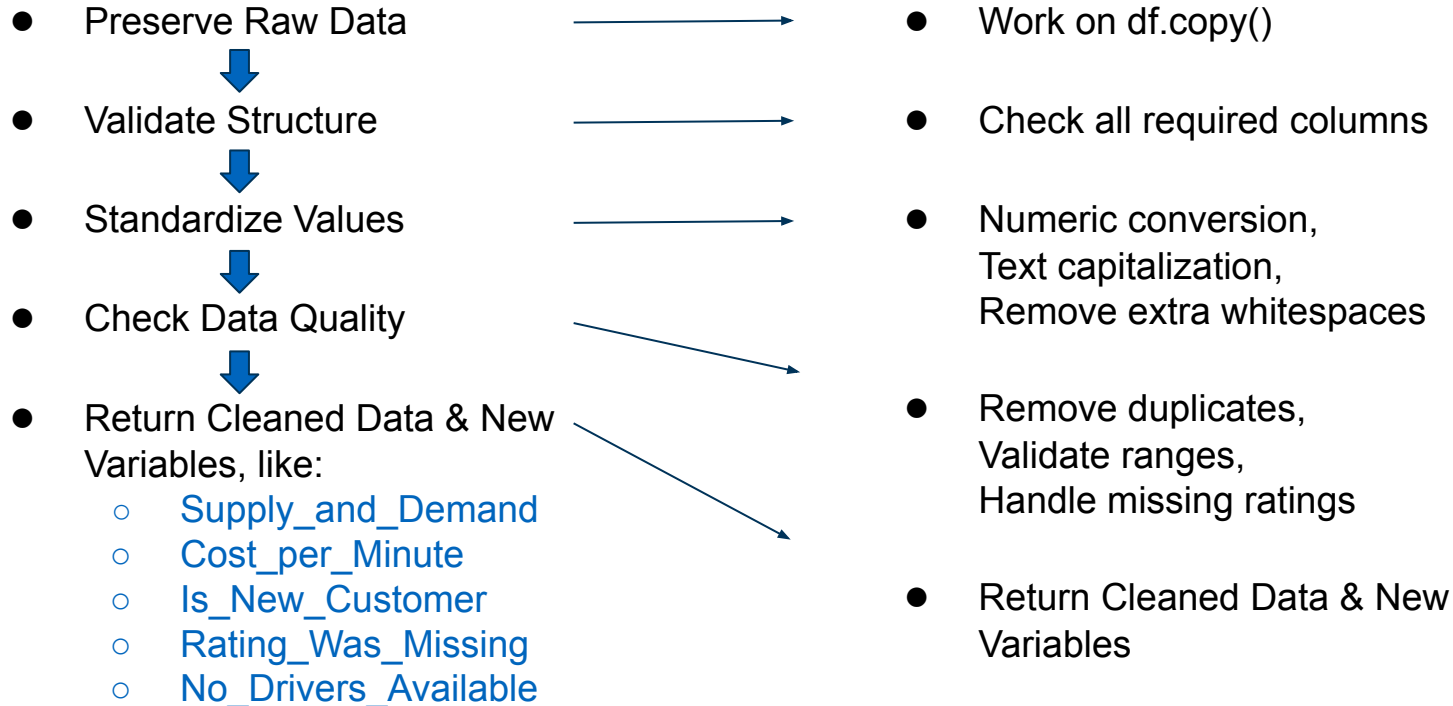
Main Variable Groups:

- Supply and Demand:
 - Number of Riders
 - Number of Drivers
- Ride Characteristics:
 - Duration
 - Vehicle Type
 - Location
 - Booking Time
- Customer Information:
 - Loyalty Status
 - Past Rides
 - Average Rating

Inspection Results:

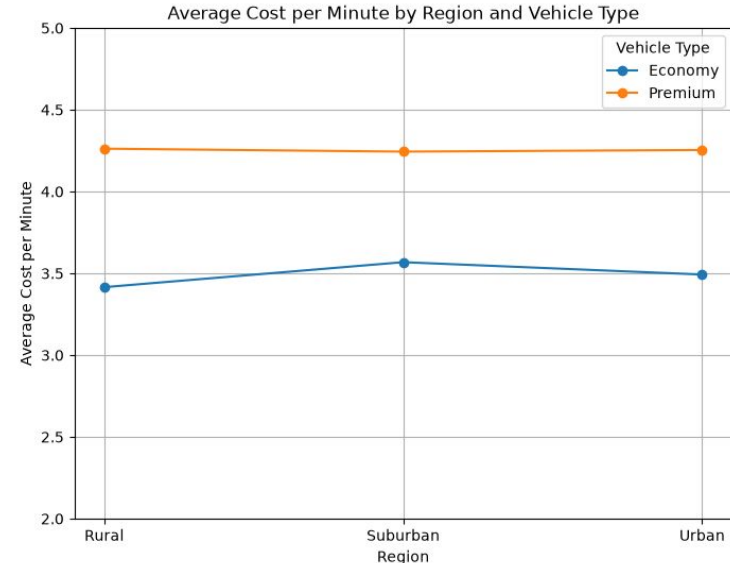
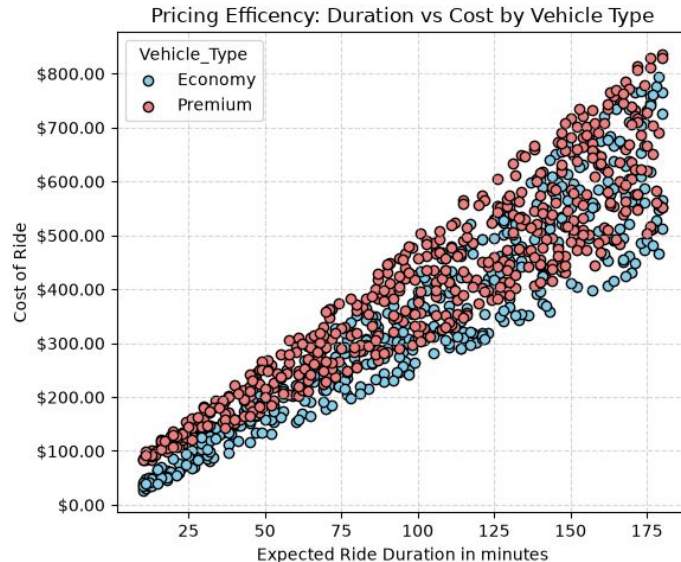
- ★ No Original Missing Values
- ★ No Duplicate Rows
- ★ Expected Categories Only
- ★ Numerical Values Stored Correctly
- ★ No obvious impossible values
- ★ Data cleaning function supports future data, more than fixing current issues

Cleaning Function



Findings: Operational View

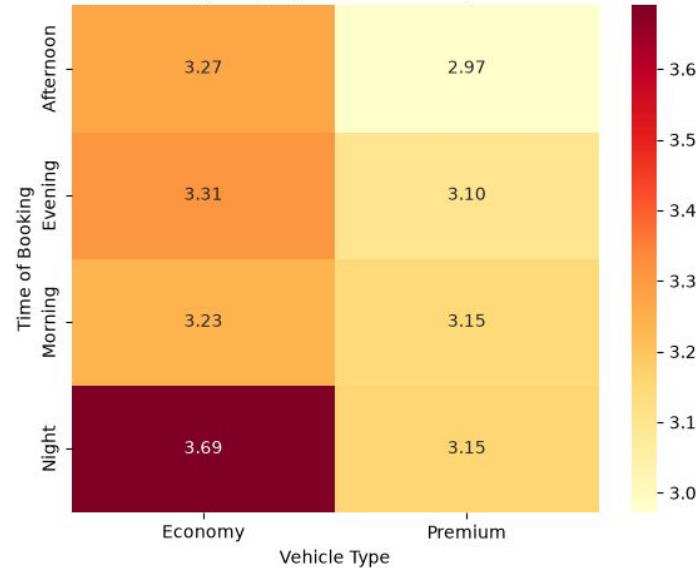
- Price grows linear with time in short rides
 - Both vehicles grow almost the same
 - In extremely long rides the prices fan out
→ long rides more vulnerable to external variables
- Minimal difference → both use same formulas with Premium with a higher base



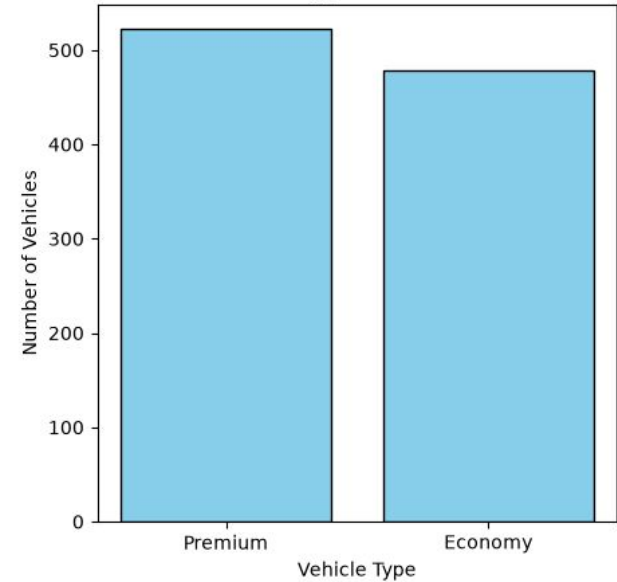
Findings: Operational View

- Overall more demand of economic vehicles
- The demand peaks at night
- Economic has less vehicles than premium → less availability

Heatmap of Average Supply and Demand by Time and Vehicle



Vehicle Type Distribution



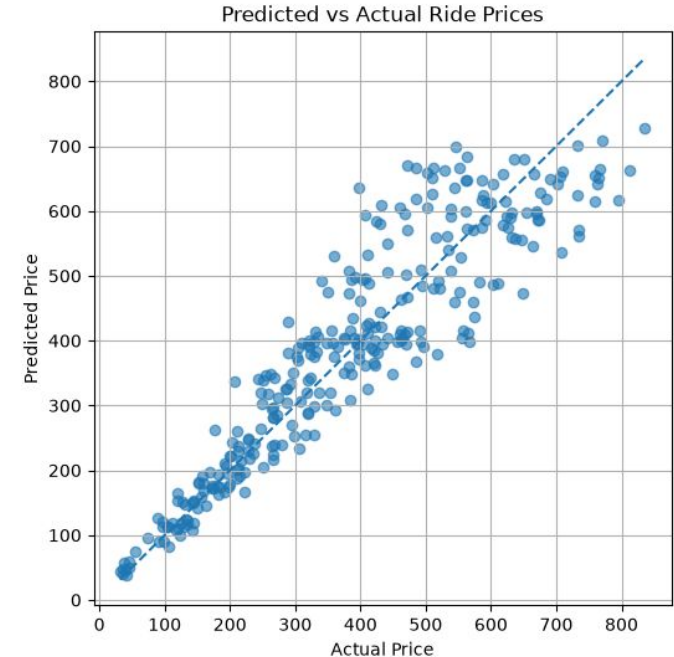
Machine Learning Model: Price Prediction

- **Goal:** Predict the historical cost of a ride
- **Model used:** Random Forest Regressor
- **Data split:** 70% training data, 30% testing data
- **Preprocessing:** One-hot encoding for categorical variables
 - Vehicle Type
 - Location Category
 - Time of Booking
 - Customer Loyalty Status
- **Target variable:** `Historical_Cost_of_Ride`

Model Evaluation Result

Evaluation Metrics	Result
R^2 Score (R^2)	0.8468
Mean Absolute Error (MAE)	\$55.52
Root Mean Squared Error (RMSE)	\$73.51

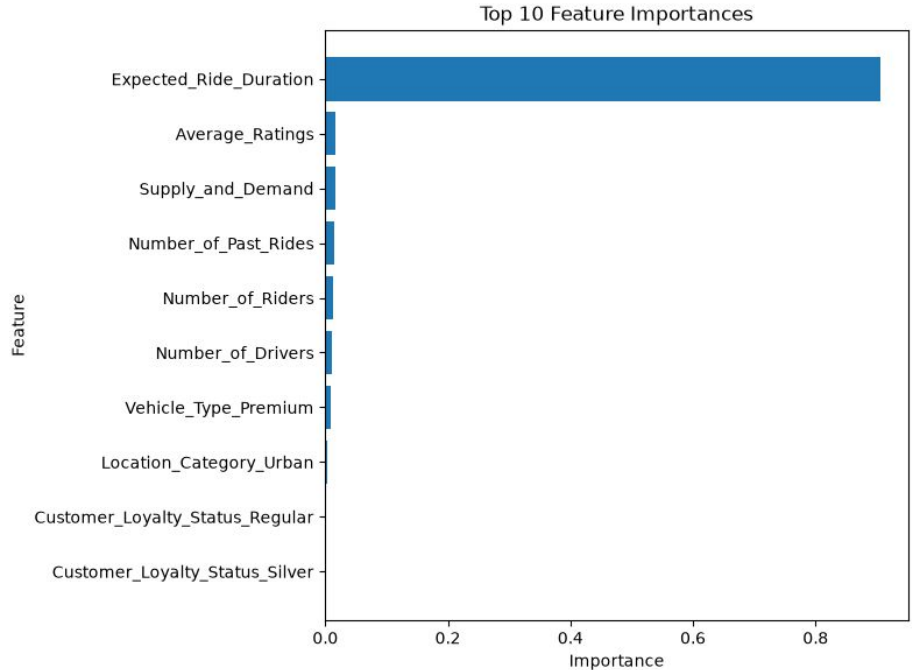
- The model explains 84.68% of the variation in ride prices.
- Most predictions follow the overall trend of the actual prices.
- Larger prediction errors mainly occur for higher-priced rides.



Feature Importance

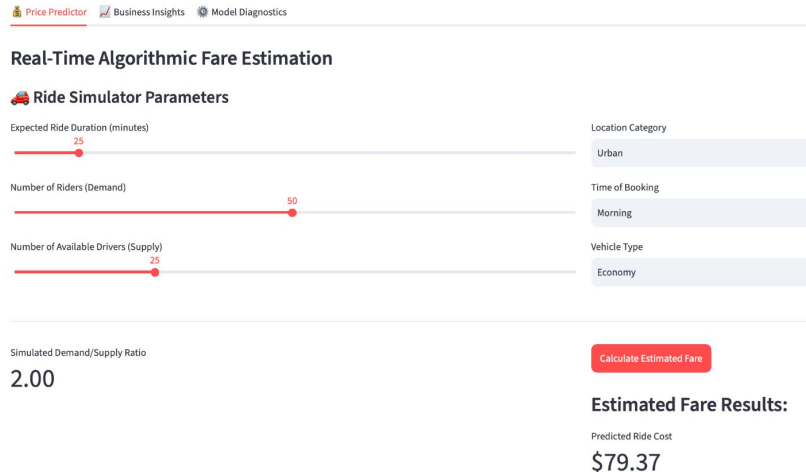
Main findings:

- Expected ride duration: ~90.6% importance
- Ratings and supply–demand: secondary factors
- Other features: limited additional impact



Business Value & The Dashboard Tool

- **Interactive Tool:** Streamlit dashboard empowers management to simulate market conditions and pricing in real-time.
- **High Confidence:** R^2 of 85% means the model is highly reliable for deployment.
- **Acceptable Error Margin:** MAE of ~\$55 (only ~15% error on an average \$372 ride).



Ride-Sharing Dynamic Pricing Dashboard

Interactive tool for exploration and real-time algorithmic price prediction.

[Price Predictor](#) [Business Insights](#) [Model Diagnostics](#)

Random Forest Model Evaluation

R^2 Accuracy Score

0.85

Mean Absolute Error (MAE)

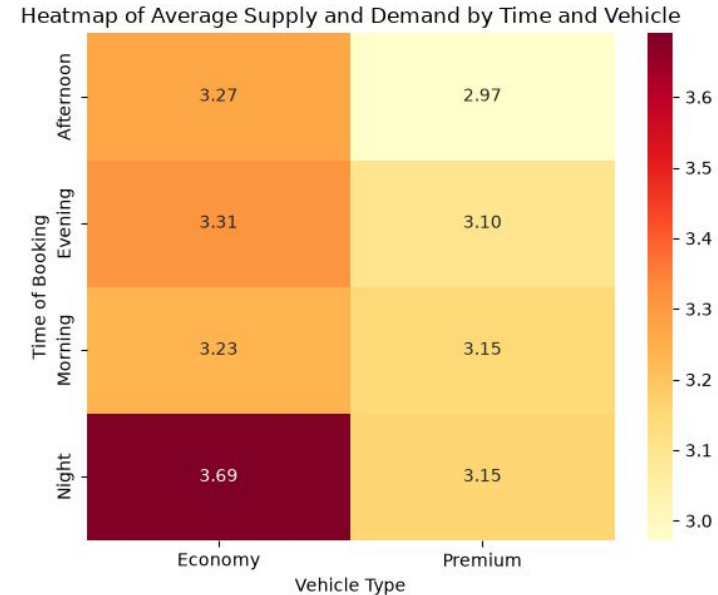
\$56

Root Mean Squared Error (RMSE)

\$74

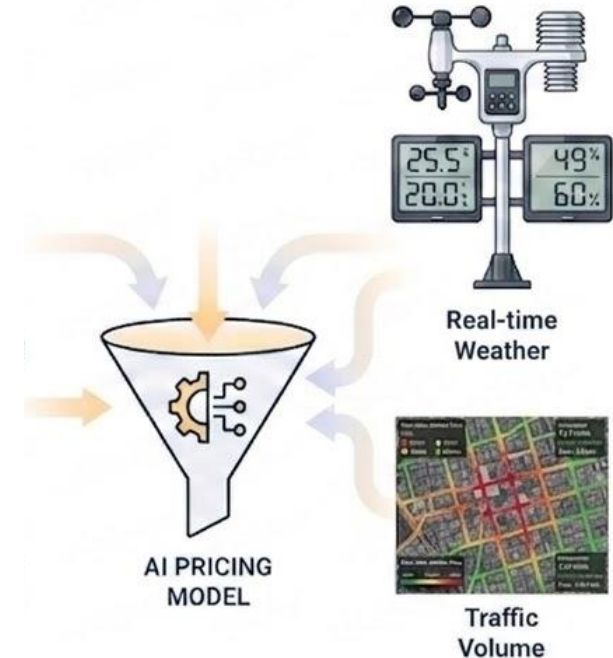
Recommendations

- **Critical Operational Imbalance:**
 - Extreme pressure on Economy cars at night (Demand/Supply ratio 3.69).
- **Incentivize Economy Drivers:**
 - Increase base pay for Economy drivers using the algorithm during night shifts.
- **Dynamic Premium Discounting:**
 - Discount Premium vehicles to safely absorb excess Economy demand without losing customers.
- **Capacity Alignment:**
 - Increase the overall baseline supply of Economy cars during night hours.



Conclusion & Future Improvements

- **Reached:**
 - Successful transition from static to data-driven algorithmic pricing strategy
- **Current Limitations:**
 - The model currently relies only on internal metrics, not the outside conditions.
- **Future Improvements:**
 - Integrate real-time external factors (e.g., weather conditions and traffic-volume patterns) to improve accuracy.



Thank you for your attention

Team Member Contribution

Lucas Krämer: Findings and Visualizations (Operational View)

Richard Horvath: Data cleaning, bug fixing

Aijia Liu: Model evaluation

Adam Pasálek: Dashboard (frontend) and Business Implications

Zhixian Huang: Utilites optimization